

Basic Tinkercad Project Report

Blinking Two LEDs Using Arduino

Objective

To control two LEDs using an Arduino Uno by providing output signals through digital pins.

Description

This project demonstrates the basics of digital output and timing using Arduino. By programming the microcontroller, we can control the state of external components like LEDs, learning about circuit connections, resistors, and the Arduino compilation process.

Components Required

- Arduino Uno
- Breadboard
- 2 × LEDs
- 2 × Resistors (220Ω or 330Ω)
- Jumper wires
- 9V Battery or USB power

Basic Concept

Arduino digital pins give HIGH (5V) or LOW (0V).
HIGH → LED ON
LOW → LED OFF
Resistor limits current and protects LED from burning out.

Pin Connections

LED 1:

Anode → Pin 8

Cathode → Resistor → GND

LED 2:

Anode → Pin 9

Cathode → Resistor → GND

Power:

Arduino powered using USB cable or 9V battery connected to VIN and GND.

Arduino Code

```
int led1 = 8;
int led2 = 9;

void setup() {
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  digitalWrite(led1, HIGH);
  digitalWrite(led2, LOW);
  delay(1000);

  digitalWrite(led1, LOW);
  digitalWrite(led2, HIGH);
  delay(1000);
}
```

Working

LED1 glows for 1 second.
LED2 glows for the next 1 second.
The LEDs blink alternatively in a continuous loop.

Output

Two LEDs blink one after another continuously.

Simulation Screenshot

